In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

In [7]:
```
df = pd.read_csv(r"C:\Users\ADMIN\Downloads\Diwali Sales Data.csv",encoding = 'u
```

In [8]:
```
df.shape
```

Out[8]: (11251, 15)

In [9]:
```
df.head()
```

Out[9]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	Stat
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtı
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Prades
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Prades
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnatak
4	1000588	Joni	P00057942	M	26-35	28	1	Gujara

In [10]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   User_ID           11251 non-null  int64  
 1   Cust_name         11251 non-null  object 
 2   Product_ID        11251 non-null  object 
 3   Gender            11251 non-null  object 
 4   Age Group         11251 non-null  object 
 5   Age               11251 non-null  int64  
 6   Marital_Status    11251 non-null  int64  
 7   State             11251 non-null  object 
 8   Zone              11251 non-null  object 
 9   Occupation        11251 non-null  object 
 10  Product_Category  11251 non-null  object 
 11  Orders            11251 non-null  int64  
 12  Amount            11239 non-null  float64
 13  Status            0 non-null      float64
 14  unnamed1          0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

In [11]:
```
df.drop(['Status','unnamed1'],axis=1, inplace = True)
```

In [12]:
```
pd.isnull(df).sum()
```

Out[12]:
```
User_ID             0
Cust_name           0
Product_ID          0
Gender              0
Age Group           0
Age                 0
Marital_Status      0
State               0
Zone                0
Occupation          0
Product_Category    0
Orders              0
Amount             12
dtype: int64
```

In [13]:
```
# drop null values
df.dropna(inplace=True)
```

In [14]:
```
#Change data type
df['Amount'] = df['Amount'].astype('int')
```

In [15]:
```
df['Amount'].dtypes
```

Out[15]:
```
dtype('int64')
```

In [16]:
```
df.columns
```

Out[16]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
       'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
       'Orders', 'Amount'],
      dtype='object')
```

In [21]:
```
# rename column
df.rename(columns = {'Marital_Status' : 'Shadi'})
```

Out[21]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Shadi	State
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat
...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra

11239 rows × 13 columns

In [22]:
```
df.describe()
```

Out[22]:

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

In [23]:
```
# Describe for specific column
df[['Age','Orders','Amount']].describe()
```

Out[23]:

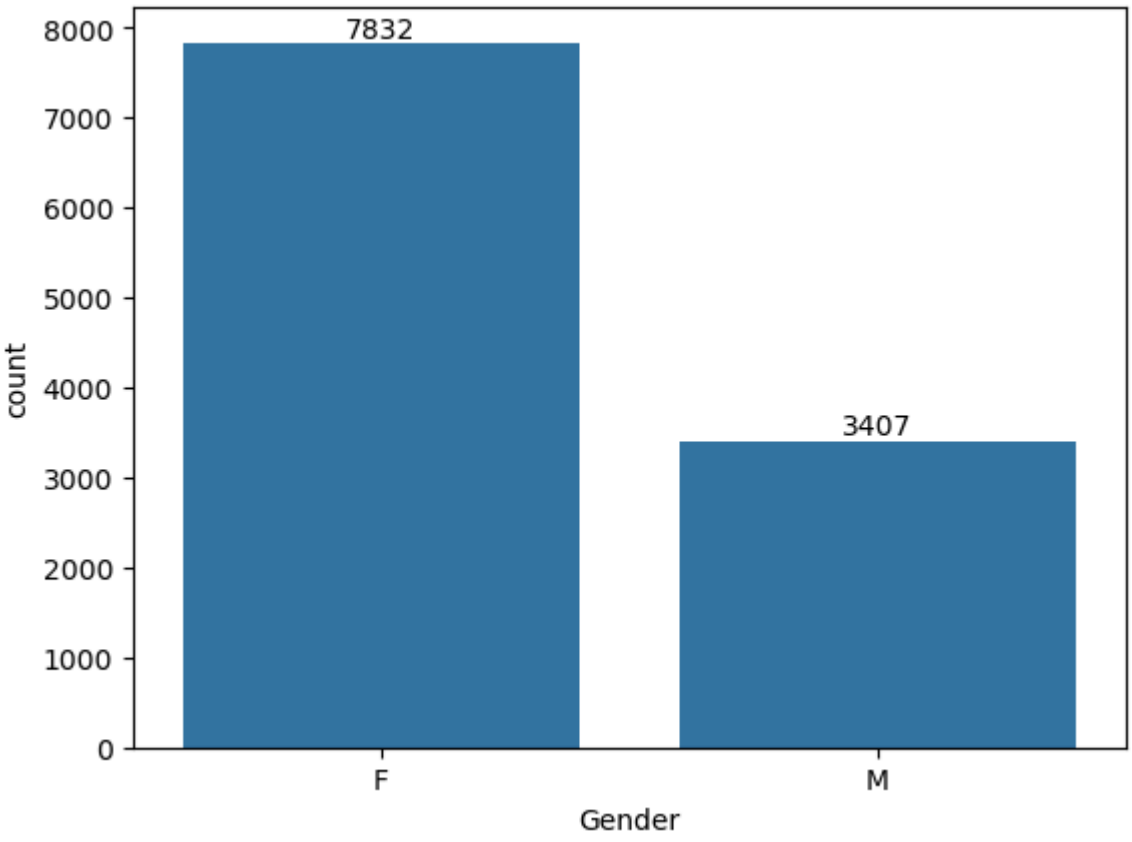
	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

Exploratory data analysis (EDA)

Gender

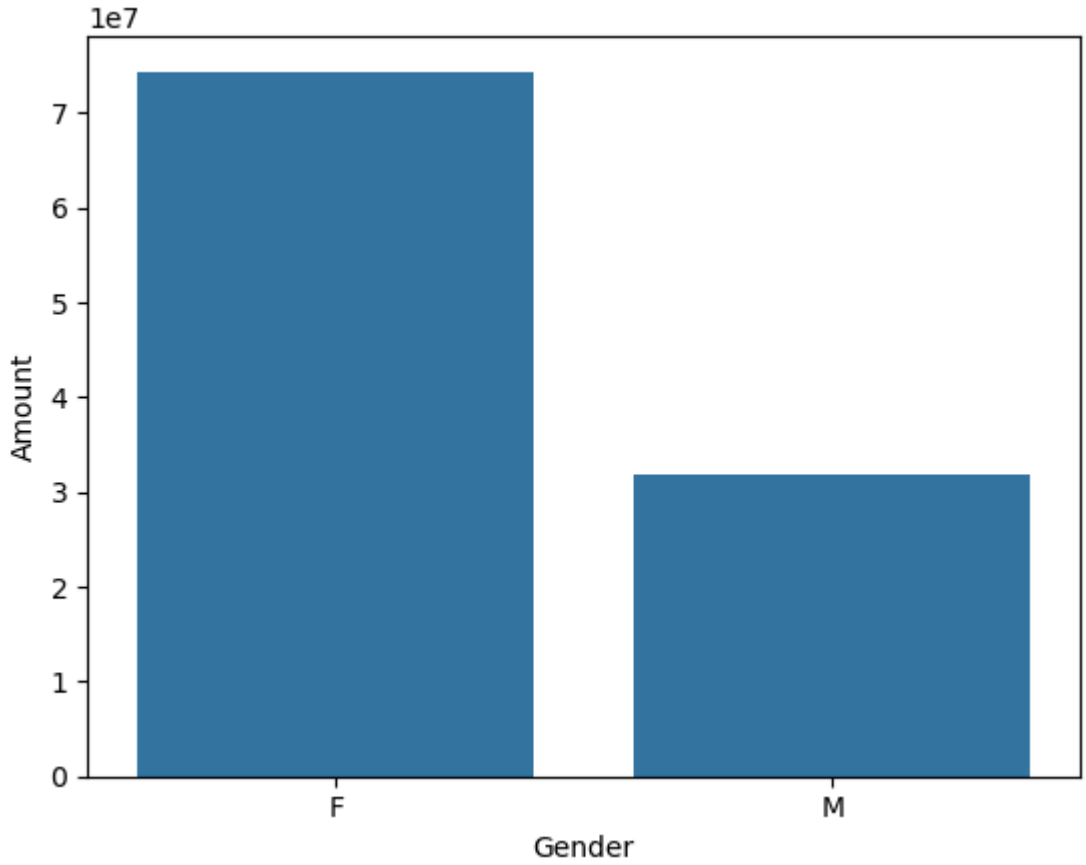
In [25]:
```
# Plotting a bar chart for gender and its count
ax = sns.countplot(x = 'Gender',data = df)

for bars in ax.containers:
    ax.bar_label(bars)
```



In [26]:
```
# Plotting a bar chart for gender vs total amount
Sales_gen = df.groupby(['Gender'],as_index = False)['Amount'].sum().sort_values(
sns.barplot(x = 'Gender' , y='Amount', data= Sales_gen)
```

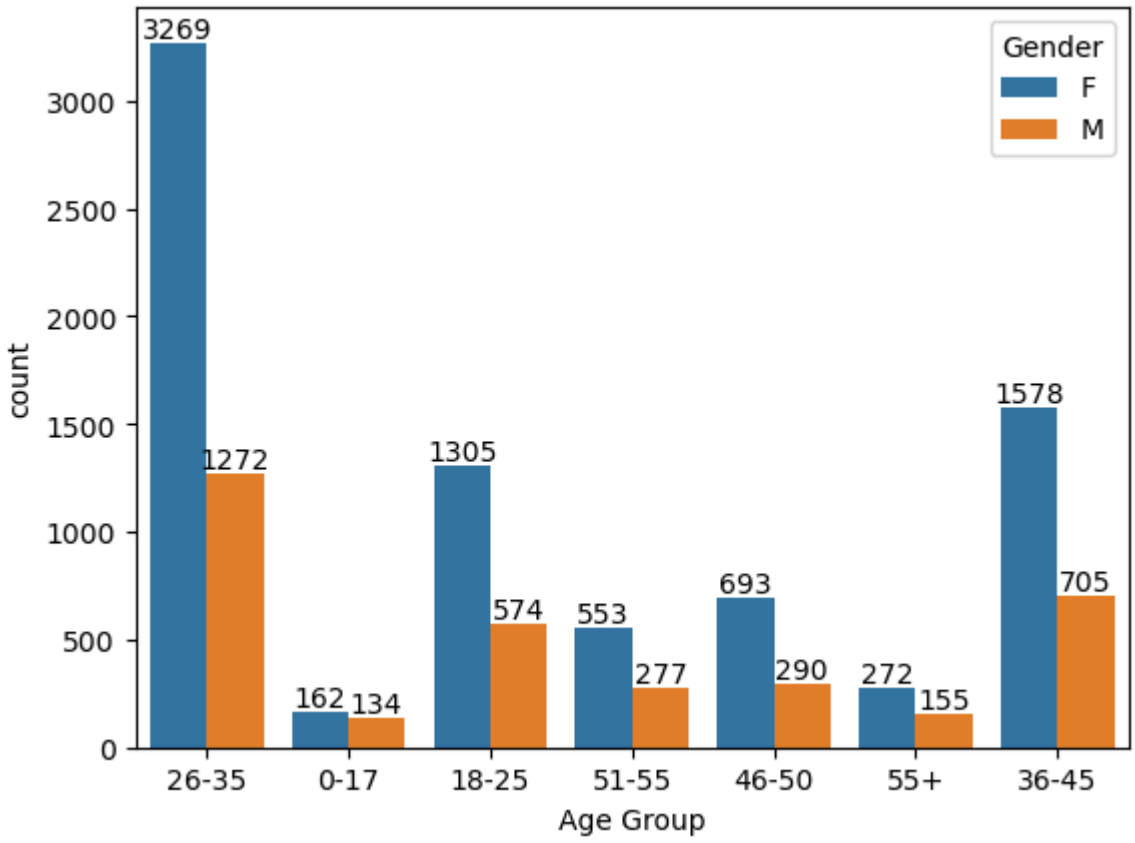
Out[26]: <Axes: xlabel='Gender', ylabel='Amount'>



Age

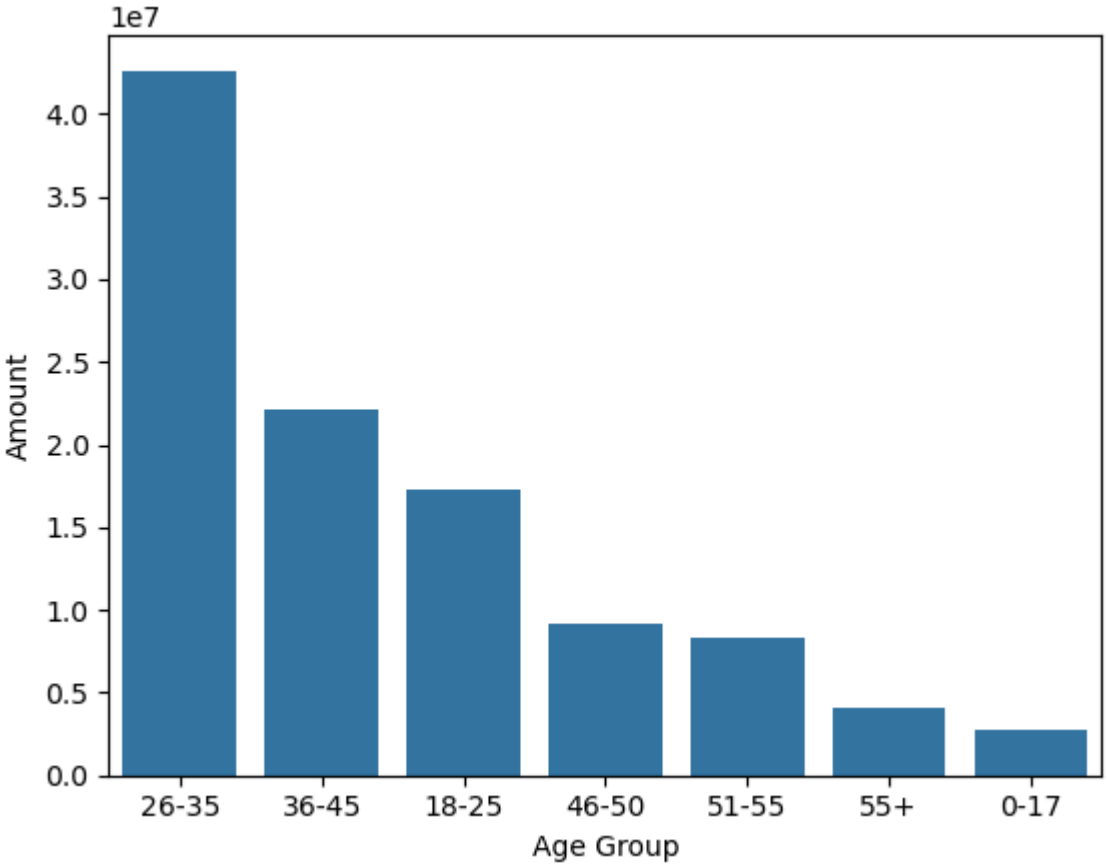
In [27]:
```
ax = sns.countplot(data=df, x = 'Age Group' , hue = 'Gender')

for bars in ax.containers:
    ax.bar_label(bars)
```



In [29]:
```
#Total amount vs age group
Sales_age = df.groupby(['Age Group'],as_index = False)['Amount'].sum().sort_valu
sns.barplot(x='Age Group',y='Amount',data=Sales_age)
```

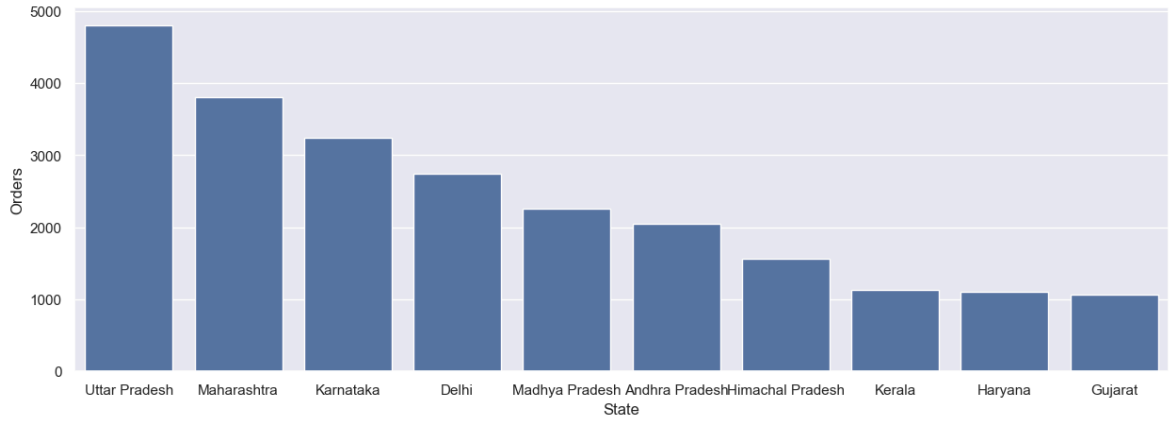
Out[29]: <Axes: xlabel='Age Group', ylabel='Amount'>



State

In [31]:
```
# Total no. of orders from top 10 states
sales_state = df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(
sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x='State',y='Orders')
```

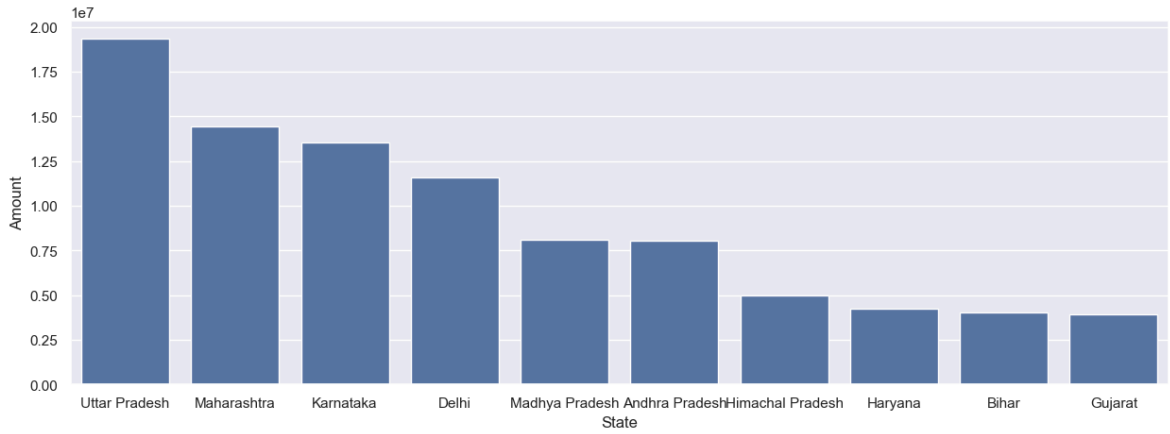
Out[31]: <Axes: xlabel='State', ylabel='Orders'>



In [32]:
```
# Total sales/amount from top 10 states
sales_state = df.groupby(['State'],as_index = False)['Amount'].sum().sort_values

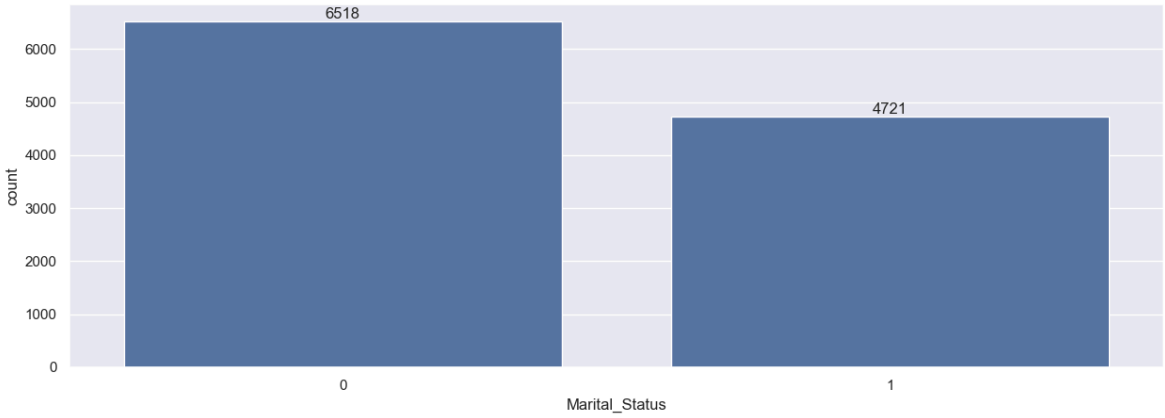
sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x='State',y='Amount')
```

Out[32]: <Axes: xlabel='State', ylabel='Amount'>



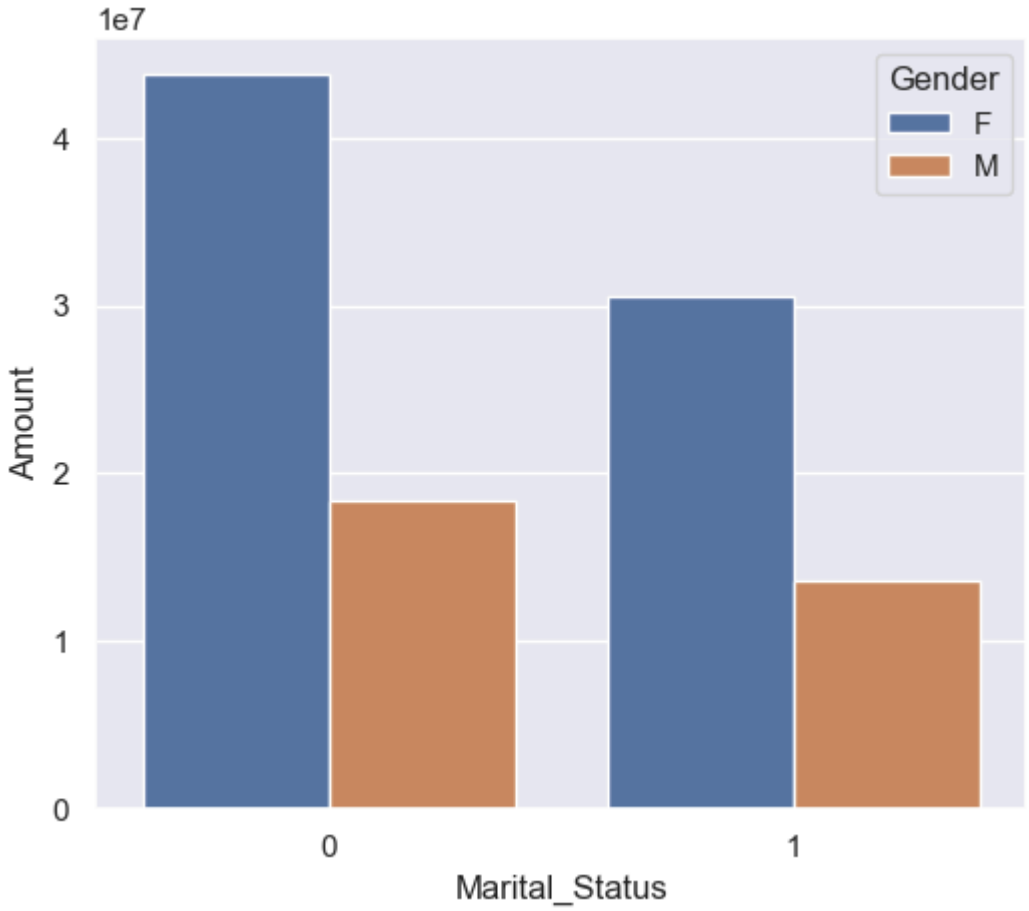
Marital status

In [33]:
```
ax = sns.countplot(data = df, x = 'Marital_Status')
sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



In [34]:
```
sales_state = df.groupby(['Marital_Status','Gender'], as_index = False)['Amount'
sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data=sales_state,x='Marital_Status',y='Amount',hue='Gender')
```

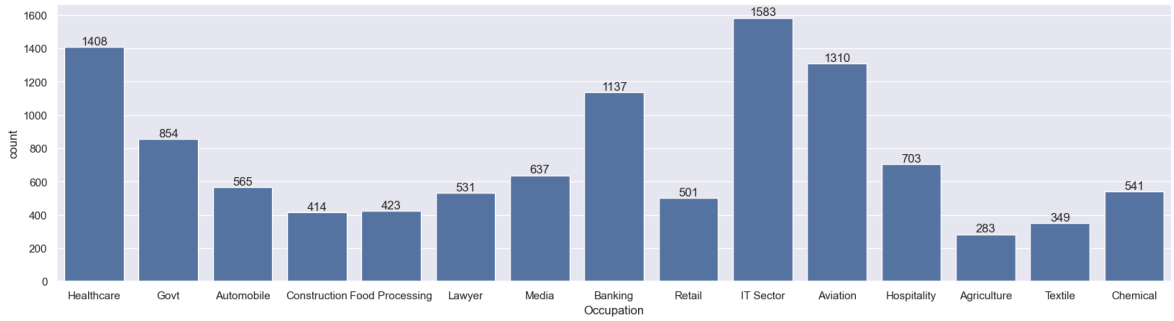
Out[34]: <Axes: xlabel='Marital_Status', ylabel='Amount'>



Occuption

In [36]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation')

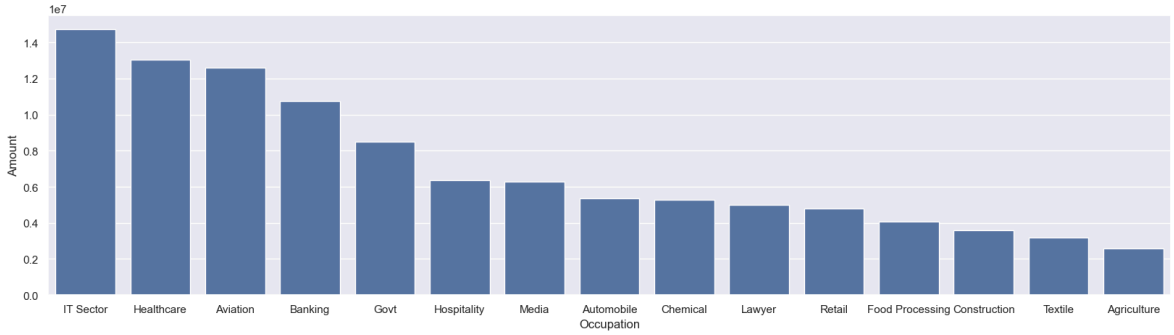
for bars in ax.containers:
    ax.bar_label(bars)
```



In [37]:
```
sales_state = df.groupby(['Occupation'], as_index=False)['Amount'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Occupation',y= 'Amount')
```

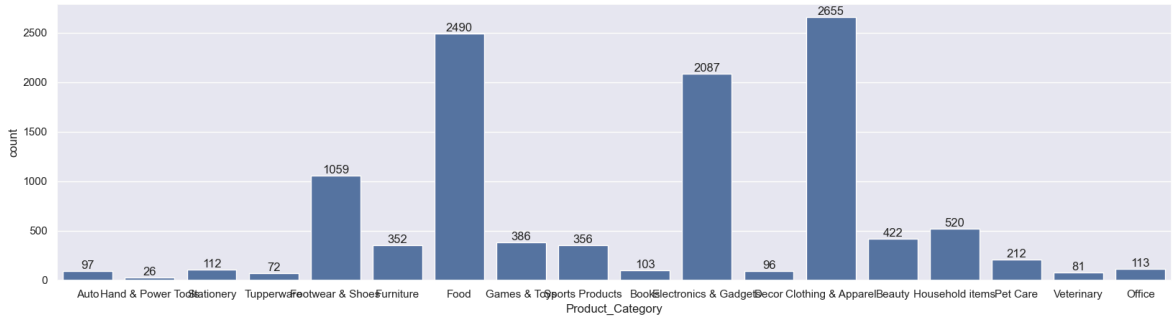
Out[37]: <Axes: xlabel='Occupation', ylabel='Amount'>



Product category

In [38]:
```
sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Product_Category')

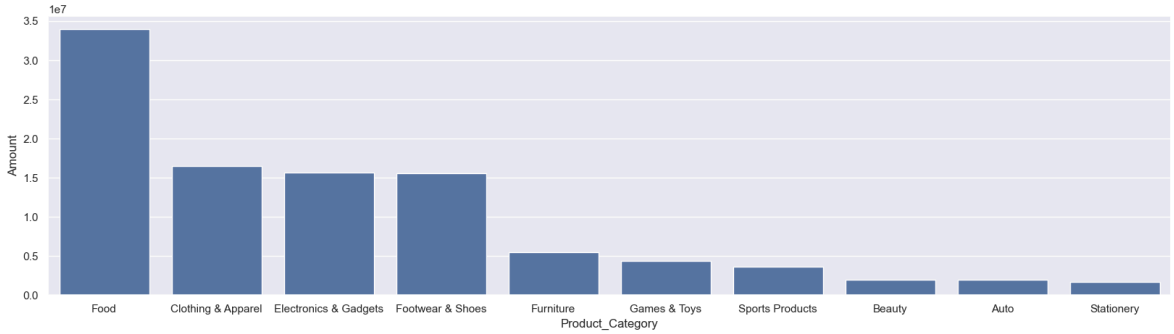
for bars in ax.containers:
    ax.bar_label(bars)
```



In [39]:
```
sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().s

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount')
```

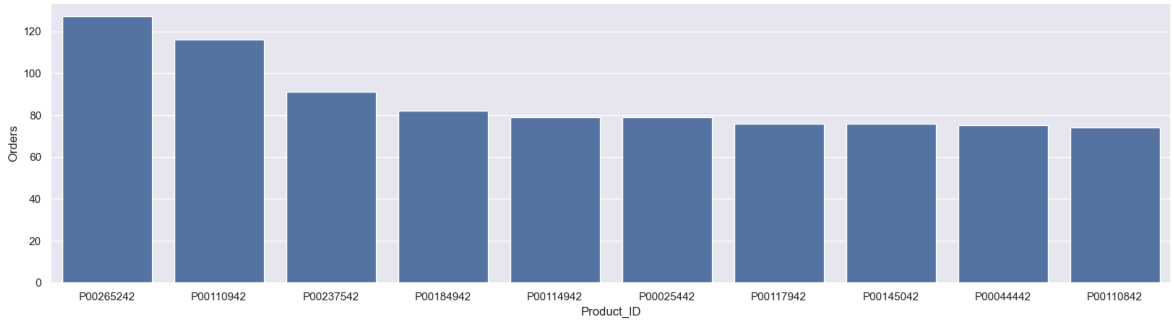
Out[39]: <Axes: xlabel='Product_Category', ylabel='Amount'>



In [40]:
```
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
```

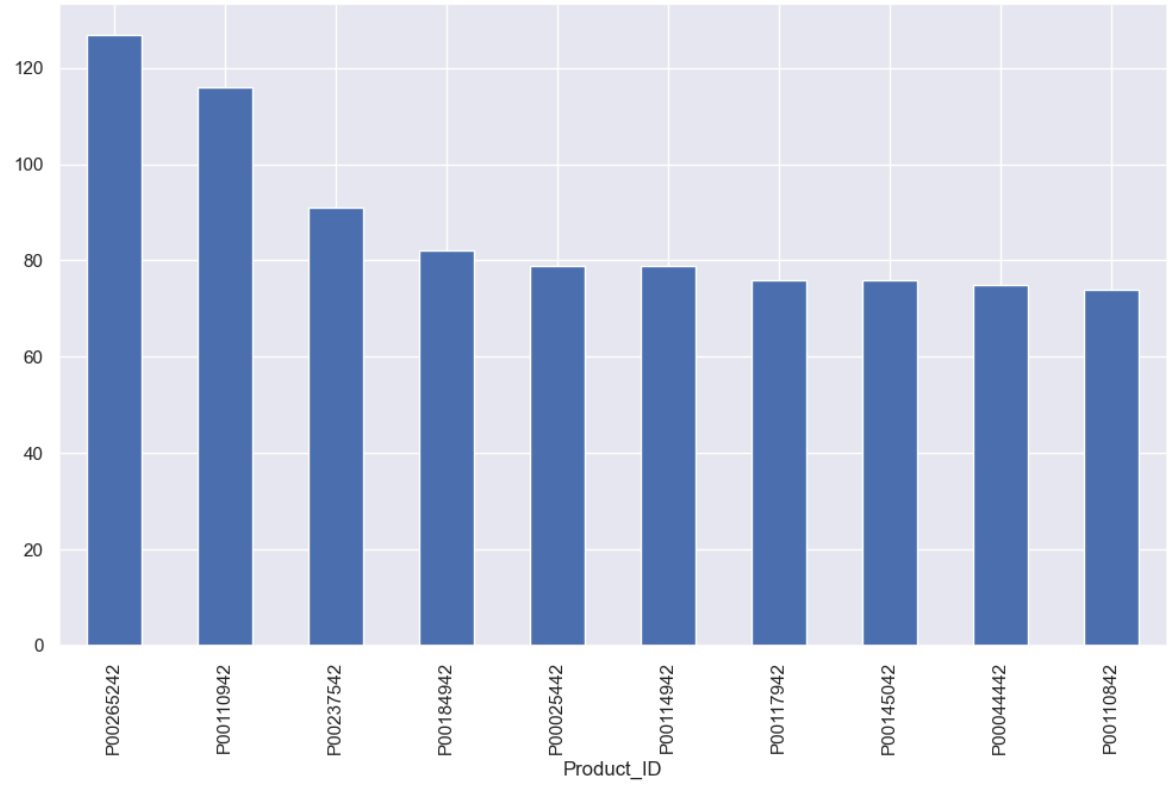
Out[40]: <Axes: xlabel='Product_ID', ylabel='Orders'>



In [41]:
```
# top 10 most sold products (same thing as above)

fig1, ax1 = plt.subplots(figsize=(12,7))
df.groupby('Product_ID')['Orders'].sum().nlargest(10).sort_values(ascending=Fals
```

Out[41]: <Axes: xlabel='Product_ID'>



In [ ]: